



# MARKERLESS GAIT ANALYSIS USING RGB-D CAMERA AND COLOR-BASED FOOT SEGMENTATION

A Procedure Presentation

Sara Cerro 189652

Coppola Manuel 189589

Silvio Carico 189661

# OVERVIEW

## Introduction

- Brief description of the problem
- target population, economic and social impact

## Proposed methodology

## Results

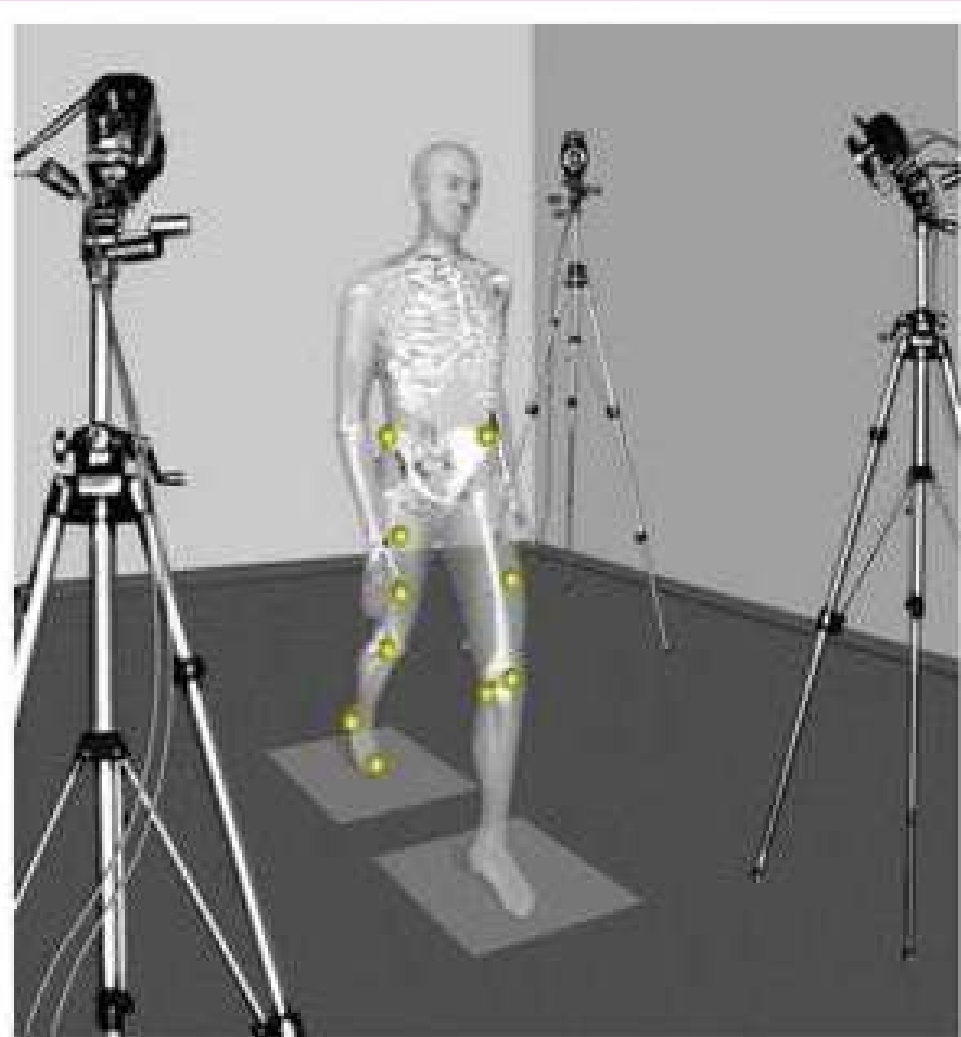
## Discussion

- Degree of technological maturity and its assessment
- Efficacy
- Drawbacks
- Future studies

## Conclusion

# INTRODUCTION

Gait analysis in biomedical engineering  
there are two main procedures to assess motor function



Marker-based systems  
reflective markers,  
expensive hardware,  
laboratory setups

Markerless-based  
systems  
RGB-D cameras and  
segmentation algorithms

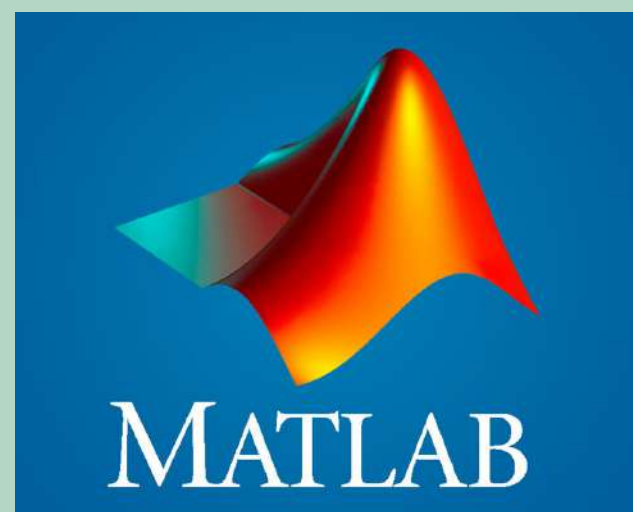
# BRIEF DESCRIPTION OF THE PROBLEM

Traditional gait laboratories are limited by high costs, the need for specialized personnel, and non-portable setups

Microsoft Kinect:



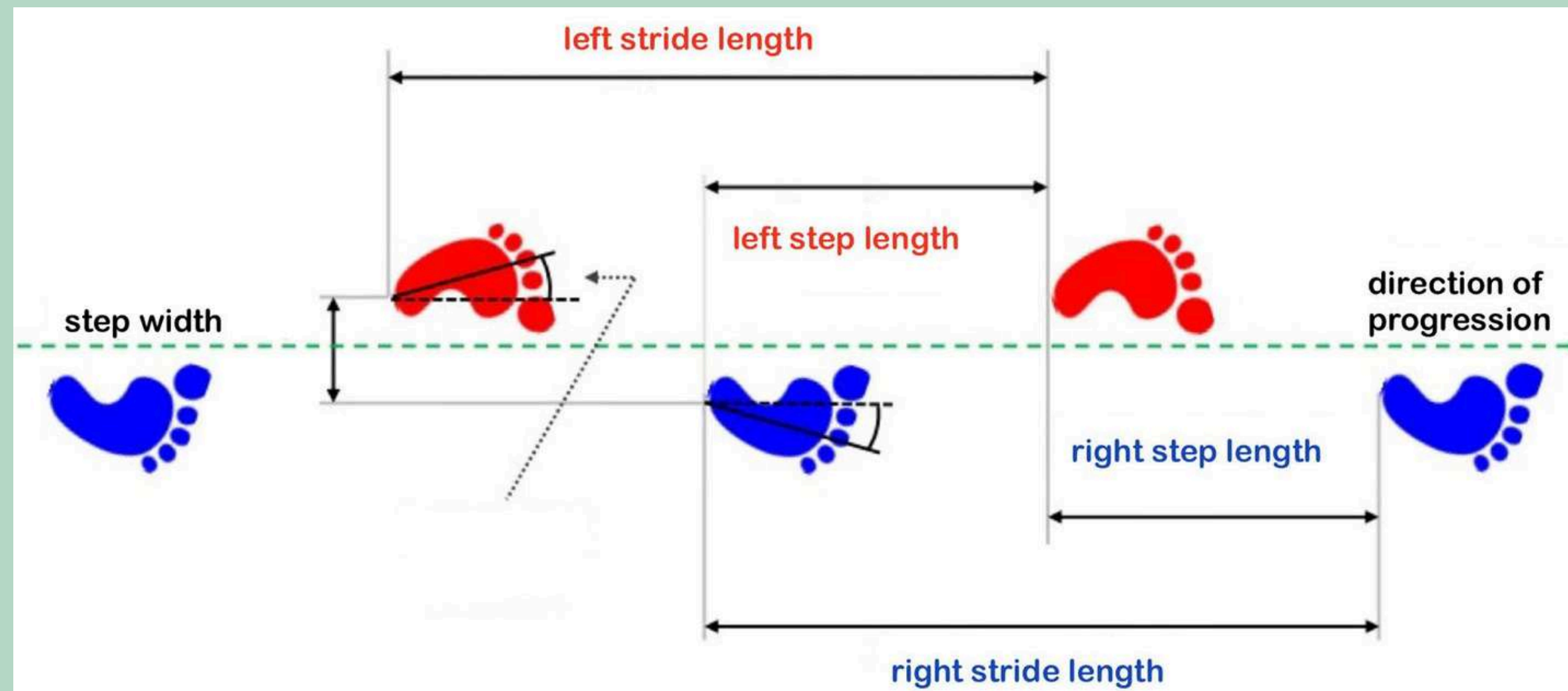
Computer vision algorithm:



This project aims to calculate key gait parameters using a low-cost, markerless system.

# BRIEF DESCRIPTION OF THE PROBLEM

Gait parameters: step length, stride length and gait speed



$$\text{Gait Speed} = \frac{\text{Stride Length (m)}}{\text{Stride Duration (s)}}$$

# TARGET POPULATION

Rehabilitation for patients with  
Neurodegenerative disorders

Elderly people at risk of falling

# SOCIAL IMPACT

Wider access to gait analysis (not just in elite labs)  
An approach more portable  
Frequent and non-intrusive monitoring prevents  
falls in elderly

# ECONOMIC IMPACT

Affordability: RGB-D Camera (\$) vs. Stereophotogrammetry (\$\$\$)

Cost Reduction: Deployment in non-laboratory settings  
Healthcare Savings: Less need for specialized personnel/facilities

# PROPOSED METHODOLOGY

TWO STAGE WORKFLOW:



Extraction of kinematic gait events from RGB frames;



Computation of spatio-temporal gait parameters.

# PROPOSED METHODOLOGY

1. Loading of the background image and application of an average filter with  $3 \times 3$  kernel

Original image:



Filtered image:



# PROPOSED METHODOLOGY

2. COLOR THRESHOLD DEFINITIONS: establishes the target RGB values for the foot markers and defines the specific tolerance ranges.

Thr = [R: 230+/- 30 G: 120+/- 35 B: 150+/- 40];

Thb = [R: 40+/- 30 G: 90+/- 30 B: 170+/- 50];

METHOD OF SELECTION: these specific values were determined empirically through a trial-and-error process.

# PROPOSED METHODOLOGY

## 3. MAIN PROCESSING LOOP:

Initialization of the binary masks:

```
FootRight = zeros(size(I,1), size(I,2));  
FootLeft = zeros(size(I,1), size(I,2));
```

Smoothing filter to the current frame+ROI:

```
hlow = fspecial('aver', n);  
I_low = imfilter(I, hlow);  
I_low(1:800, :, :) = 0;
```

Color segmentation logic to fill the masks:

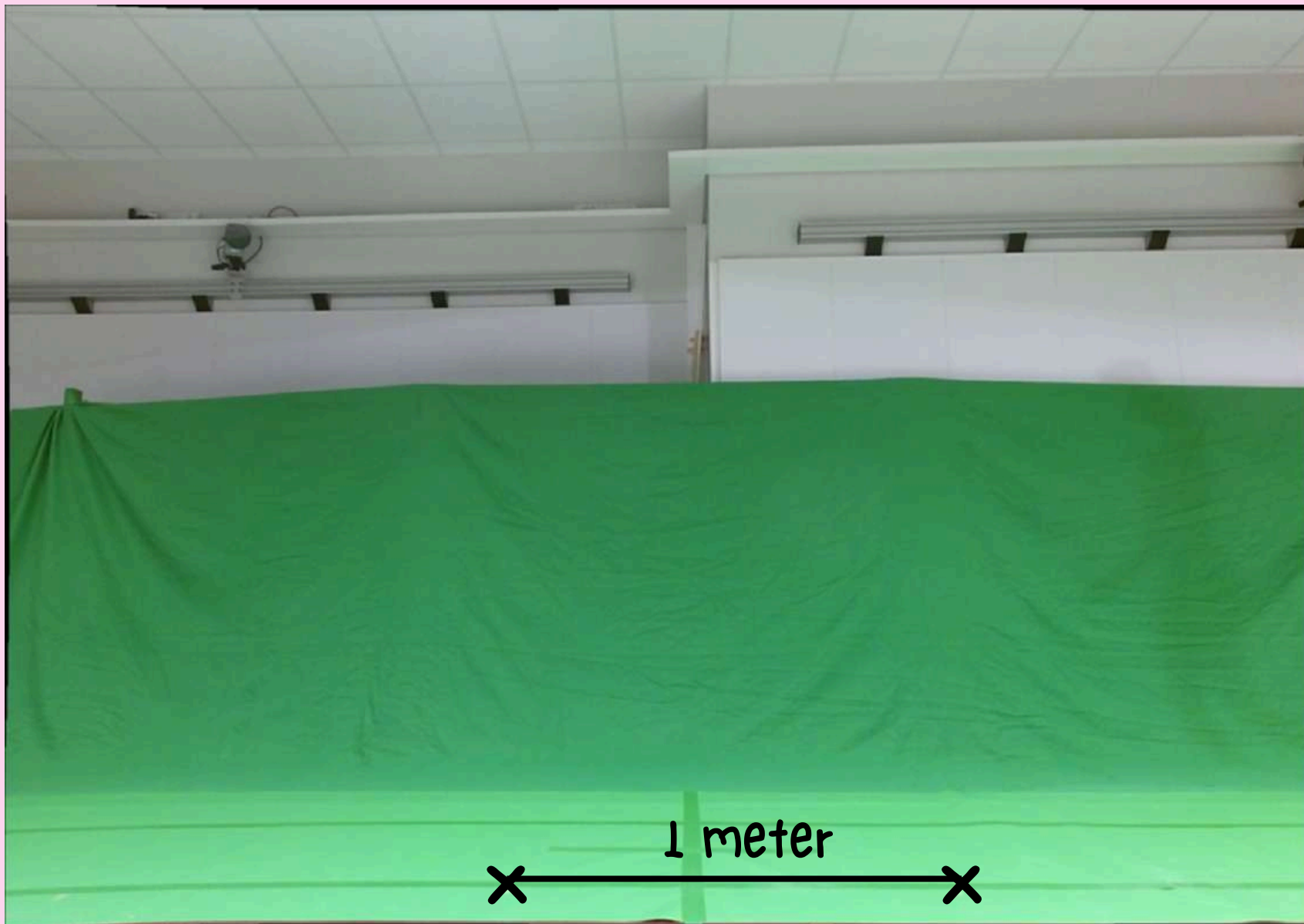
```
FootRight(i,j) = 1 or FootLeft(i,j)=1
```

Creation of two matrices CR  
and CL with the coordinates  
of the centroids

```
s_right = regionprops(FootRight, 'Centroid');  
s_left = regionprops(FootLeft, 'Centroid');  
  
CR(k-first_frame+1,:) = cat(1, s_right.Centroid);  
CL(k-first_frame+1,:) = cat(1, s_left.Centroid);
```

# PROPOSED METHODOLOGY

4. Manual calibration to find the conversion factor from pixels to meters:



```
[x,y] = ginput(2);  
distancePixel = abs(x(2)-x(1));  
Factor = 1/distancePixel;
```

# PROPOSED METHODOLOGY

## 5. GAIT EVENT DETECTION STRATEGY:

We use the horizontal distance between feet to identify gait events:

```
distance_Feet = abs(CR(:,1)-CL(:,1));
```

Finding the peaks (maximum separation in pixel):

```
[pks, locs] = findpeaks(distance_Feet);  
Max Distance = Heel Strike
```

The person start to walk with the left foot and makes 3 steps:

```
LContact = locs([1 3]);  
RContact = locs(2)
```

# PROPOSED METHODOLOGY

## 6. PARAMETER CALCULATION

(All spatial values are converted from pixels to meters using the calibration 'Factor')

- **STRIDE LENGTH:** The distance covered between two consecutive HS (heel strikes) of the *SAME* foot (e.g., Left → Left).

```
Stride_pixel = abs(CL(LContact(1), 1) - CL(LContact(2), 1));
```

- **STEP LENGTH:** The distance between the HS of one foot and the subsequent HS of the *OPPOSITE* foot (e.g., Left → Right).

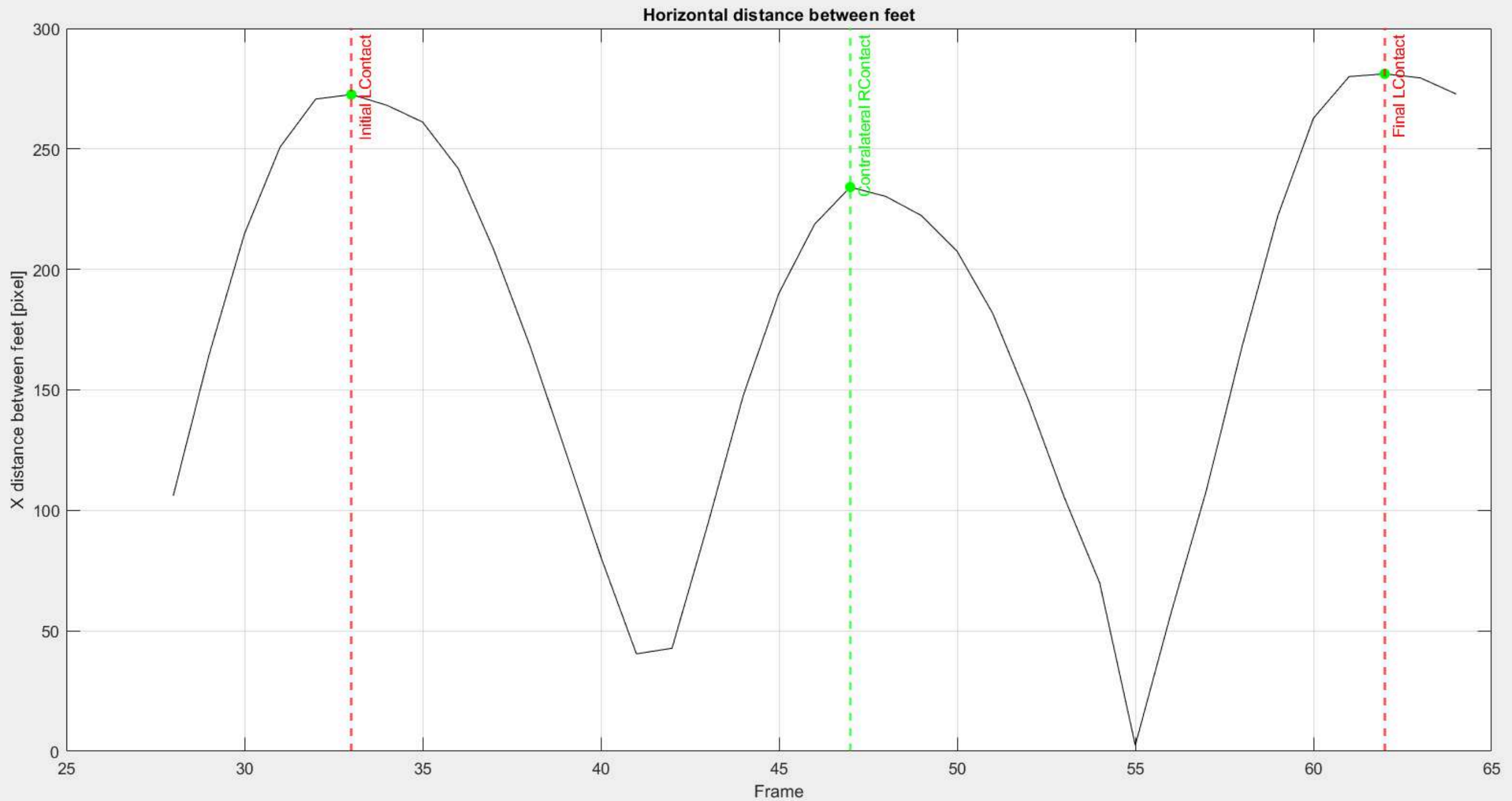
```
Step_pixel_right = abs(CL(LContact(1), 1) - CR(RContact(1), 1));  
Step_pixel_left = abs(CL(LContact(2), 1) - CR(RContact(1), 1));
```

- **GAIT SPEED:** Calculated as the Stride Length divided by the time elapsed (Stride Duration).

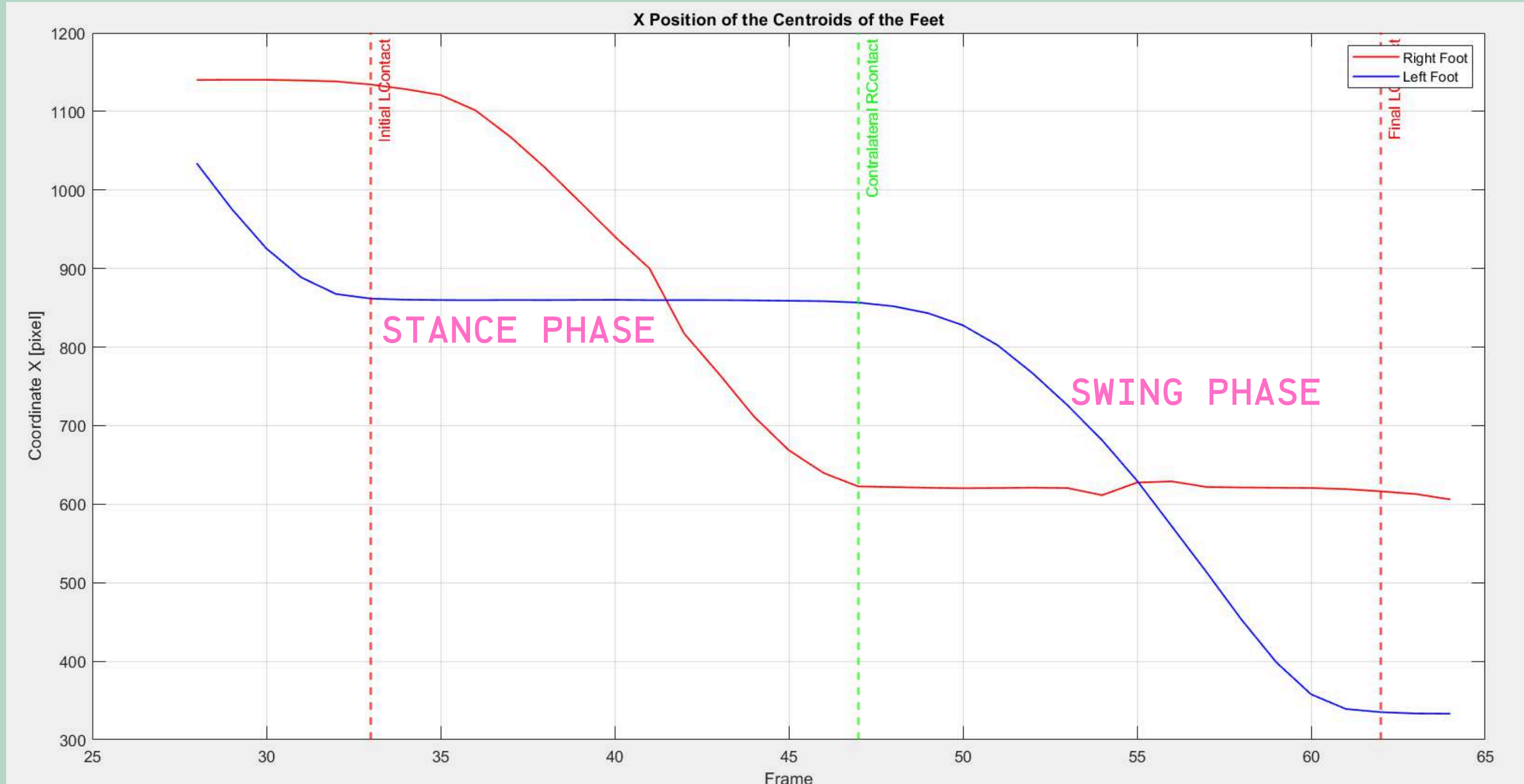
```
time = abs((LContact(2) - LContact(1))) / frame_rate;  
gait_speed = Stride_length / time;
```

# RESULTS

## Periodicity & Symmetry



# RESULTS



# RESULTS

|                                    |        |
|------------------------------------|--------|
| Conversion Factor pixel to meters: | 0,0019 |
| Stride Length (m):                 | 1,0004 |
| Step length Right (m):             | 0,4544 |
| Step Length Left (m):              | 0,5460 |
| Gait speed (m/s):                  | 1,0349 |

CONCLUSION: The extracted parameters align with normative values reported in literature for healthy individuals [6], [7], [8].

# DISCUSSION

## VISUAL VALIDATION

the system was tested on an individual with normal motor abilities, showing the expected high degree of symmetry.

the segmentation masks perfectly match the real foot positions, confirming high tracking accuracy on the healthy subject .

## SIGNAL ROBUSTNESS

the inter-foot distance signal displayed a stable, periodic waveform across all gait cycles.

the centroid trajectories are smooth and periodic. The inter-foot distance peaks align precisely with the initial contact events .

## PHYSIOLOGICAL CONSISTENCY

the extracted spatio-temporal parameters (stride length, gait speed) fall within the normative ranges for healthy individuals.

# DEGREE OF TECHNOLOGICAL MATURITY AND ITS ASSESSMENT

The system is a working prototype based on established computer vision techniques, successfully validated in a lab environment.

Maturity is currently constrained to controlled experimental conditions (uniform lighting, single subject). Robustness in uncontrolled "wild" scenarios has not yet been assessed.



# EFFICACY

The pipeline reliably identifies feet across all frames without losing the target, even with simple color thresholding.

The periodicity of the distance signal allows for accurate detection of gait events without false positives.

Proves that meaningful clinical metrics can be obtained with low-cost hardware and minimal computational complexity.



# DRAWBACKS

**Environmental Sensitivity:** Accuracy strongly depends on controlled illumination and distinct color contrast (specific socks); lighting deviations or reflections compromise segmentation.

**Limited Generalizability:** Validated only on a single healthy subject; robustness on pathological gait or atypical movement patterns remains untested.

**Coarse Kinematics:** Tracking centroid positions provides only an approximation, missing detailed biomechanics like joint angles or precise foot orientation.

**Scale Inaccuracies:** Reliance on a fixed, manually derived calibration factor introduces potential measurement errors.

# FUTURE STUDIES

- Integrate Machine Learning or Deep Learning classifiers to reduce sensitivity to lighting and increase reliability in diverse environments.
- Test on a broader population including different anthropometrics, footwear, and pathological gait patterns to ensure generalizability.
- Incorporate multi-view setups or depth information to improve spatial accuracy and extract richer kinematic parameters.

# CONCLUSIONS

- We successfully demonstrated that a single RGB camera can extract reliable gait parameters.
- The system provides a valid, portable, and low-cost alternative to cumbersome marker-based setups.
- Despite current limitations, the methodological core is robust and ready for future clinical development.